OFFICIAL TECHNICAL SPECIFICATION DATASHEET
MANUFACTURER: FIRST SOLAR
MODEL / SERIES: Series 6 Plus FS-6460
DOCUMENT TITLE: First Solar Series 6 Plus Photovoltaic Module Technical Datasheet
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1. PRODUCT IDENTIFICATION & ARCHITECTURE
  Cell Technology: Thin-film CdTe
  Module Construction: Monofacial
  Number of Cells / Format:Thin-film Cadmium Telluride (CdTe) continuous semiconductor layer
  Glass / Front Encapsulant:Heat-strengthened front glass laminated to heat-strengthened back
  Frame Material: Anodized Aluminium with Under-mount Mounting Rails
  Junction Box / Protection: IP68 with bypass diodes

2. ELECTRICAL CHARACTERISTICS AT STC (1000 W/m2, 25C, AM 1.5)
  Nominal Maximum Power (Pmax / Pnom): 460.0 Wp
  Module Aperture Efficiency: 18.60 %
  Open Circuit Voltage (Voc): 222.90 V
  Short Circuit Current (Isc): 2.59 A
  Maximum Power Voltage (Vmp): 188.80 V
  Maximum Power Current (Imp): 2.44 A
  Maximum System Voltage: 1500.0 V DC
  Maximum Series Fuse Rating: 5.0 A

3. TEMPERATURE CHARACTERISTICS & OPERATING LIMITS
  Temp. Coefficient of Pmax (gamma): -0.320 %/C
  Temp. Coefficient of Voc (beta): -0.280 %/C
  Temp. Coefficient of Isc (alpha): +0.040 %/C
  Nominal Cell Temperature (NOCT/NMOT): 45 C (NMOT)
  Operating Temperature Range: -40 C to +85 C

4. MECHANICAL DIMENSIONS & RATINGS
  Length: 2009.0 mm
  Width: 1232.0 mm
  Thickness (Height): 49.0 mm
  Gross Aperture Area: 2.4751 m2
  Module Weight: 34.5 kg
  Mechanical Design Load (Front): 5400 Pa (Snow load)
  Mechanical Design Load (Rear): 2400 Pa (Wind load)

5. WARRANTY & LIFETIME PERFORMANCE
  Product Workmanship Warranty: 12.0 Years
  Linear Power Output Warranty: 25.0 Years
  Annual Linear Degradation: 0.50 % / year
  End of Warranty Retained Output: 85.0 %

6. PROVENANCE & ARCHIVAL METADATA
  Authoritative URL: https://www.firstsolar.com/-/media/First-Solar/Technical-Documents/Serie
  Document Verification: OFFICIAL_FAMILY_DOCUMENT
  Installation Mapping: BENCHMARK_REFERENCE_MODULE
  Verification Date: 2026-11-10

SolarSelect MLOps Hardware Database Reference Standard IEC 61215 / IEC 61730